

Session 16 (AIML) -Optional Task

- **Name of Student :** Shruti Dnyaneshwar Darwatkar
- **Domain Name:** AIML Development
- **College Name:** Zeal Polytechnic , Narhe
- **Branch Name:** AIML
- **Github url:** https://github.com/shrutidarwatkar-sketch/Task_ITR.git

Q1. Load the patients dataset using pandas and display the first 10 rows of the data.

Program screenshot:

The screenshot shows a Google Colab environment. In the first cell, a file named 'patients.csv' is uploaded. The second cell contains the following Python code:

```
import pandas as pd
patients= pd.read_csv("patients.csv")
patients.head(10)
```

The output of the code is a table showing the first 10 rows of the dataset:

	patient_id	name	age	arrival_date	departure_date	service	satisfaction
0	PAT-09484753	Richard Rodriguez	24	2025-03-16	2025-03-22	surgery	61
1	PAT-f0644084	Shannon Walker	6	2025-12-13	2025-12-14	surgery	83
2	PAT-ac6162e4	Julia Torres	24	2025-06-29	2025-07-05	general_medicine	83
3	PAT-3dda2bb5	Crystal Johnson	32	2025-10-12	2025-10-23	emergency	81
4	PAT-08591375	Garrett Lin	25	2025-02-18	2025-02-25	ICU	76
5	PAT-f4b29bae	Diana May	83	2025-06-26	2025-06-30	emergency	81
6	PAT-283cda07	William Herrera	62	2025-12-26	2025-12-27	emergency	66
7	PAT-5b61868c	Ashley Waller	0	2025-05-21	2025-06-04	ICU	82
8	PAT-f9c8afa6	Victor Baker	50	2025-07-30	2025-08-13	general_medicine	91
9	PAT-5290be70	Jeffrey Chandler	29	2025-11-01	2025-11-14	emergency	88

Q2. Use pandas functions to get basic information about the dataset:

- Shape of the data (shape)
- Column names and data types (info())
- Statistical summary of numerical columns (describe())

Program screenshot:

```
import pandas as pd
#load the dataset
patients= pd.read_csv("patients.csv")

#1.shape of the dataset
print("shape of the dataset")
print(patients.shape)

#2. column names and data types
print("\n dataset information")
patients.info()

#3. Statistical summary of numerical columns
print("\n statistical summary")
print(patients.describe())
```

```
... shape of the dataset
(1000, 7)

dataset information
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   patient_id            1000 non-null   object
1   name                  1000 non-null   object
2   age                   1000 non-null   int64
3   arrival_date          1000 non-null   object
4   departure_date        1000 non-null   object
5   service               1000 non-null   object
6   satisfaction           1000 non-null   int64
dtypes: int64(2), object(5)
memory usage: 54.8+ KB

statistical summary
      age  satisfaction
count  1000.000000    1000.000000
mean    45.337000     79.597000
std     25.999912     11.550325
min       0.000000     60.000000
25%     23.000000     70.000000
50%     46.000000     80.000000
75%     68.000000     89.250000
max     89.000000     99.000000
```

Q3. Check if the dataset contains any null/missing values. Show the count of null values for each column.

Program screenshot:

```
#check missing values
print("missing values in each column:")
print(patients.isnull().sum())

... missing values in each column:
patient_id      0
name            0
age             0
arrival_date    0
departure_date  0
service         0
satisfaction    0
dtype: int64
```

Q4. Identify numerical and categorical columns in the dataset. Print them separately.

Program screenshot:

```
# Numerical columns
numerical_cols = patients.select_dtypes(include=['int64', 'float64']).columns
print("Numerical columns:")
print(numerical_cols)

# Categorical columns
categorical_cols = patients.select_dtypes(include=['object']).columns
print("\n Categorical columns:")
print(categorical_cols)

Numerical columns:
Index(['age', 'satisfaction'], dtype='object')

Categorical columns:
Index(['patient_id', 'name', 'arrival_date', 'departure_date', 'service'], dtype='object')
```

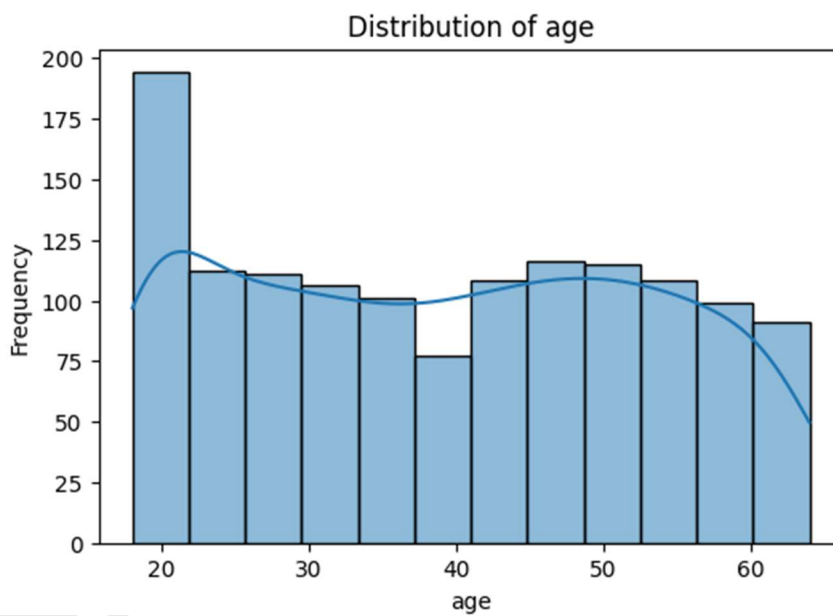
Q5. Create distribution plots for all numerical columns (age, satisfaction).

Program screenshot:

```
import matplotlib.pyplot as plt
import seaborn as sns

# List of numerical columns
num_cols = ['age', 'bmi', 'children', 'charger']
for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(insurance[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```

...

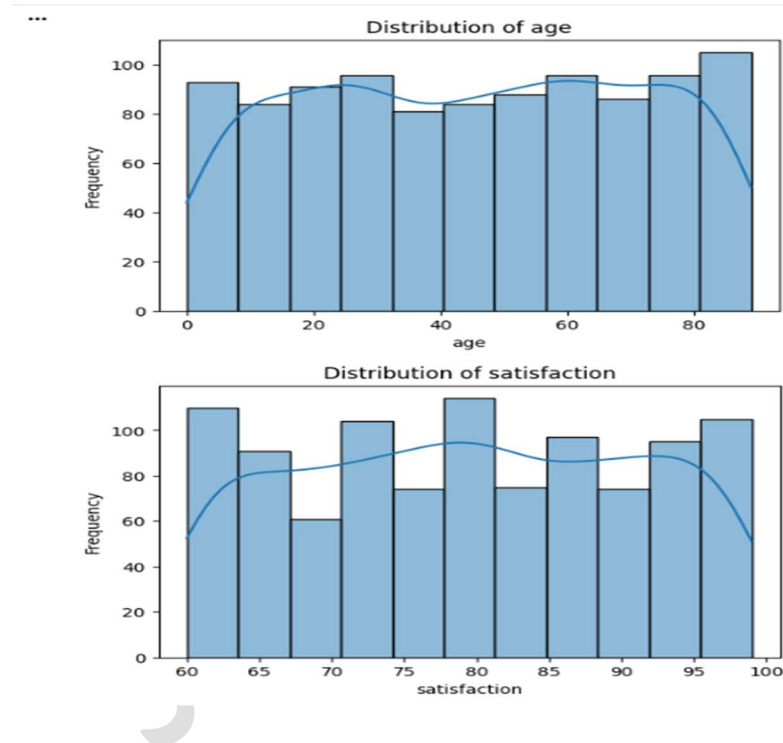


Q6. Create count plots for all categorical columns (service)

Program screenshot:

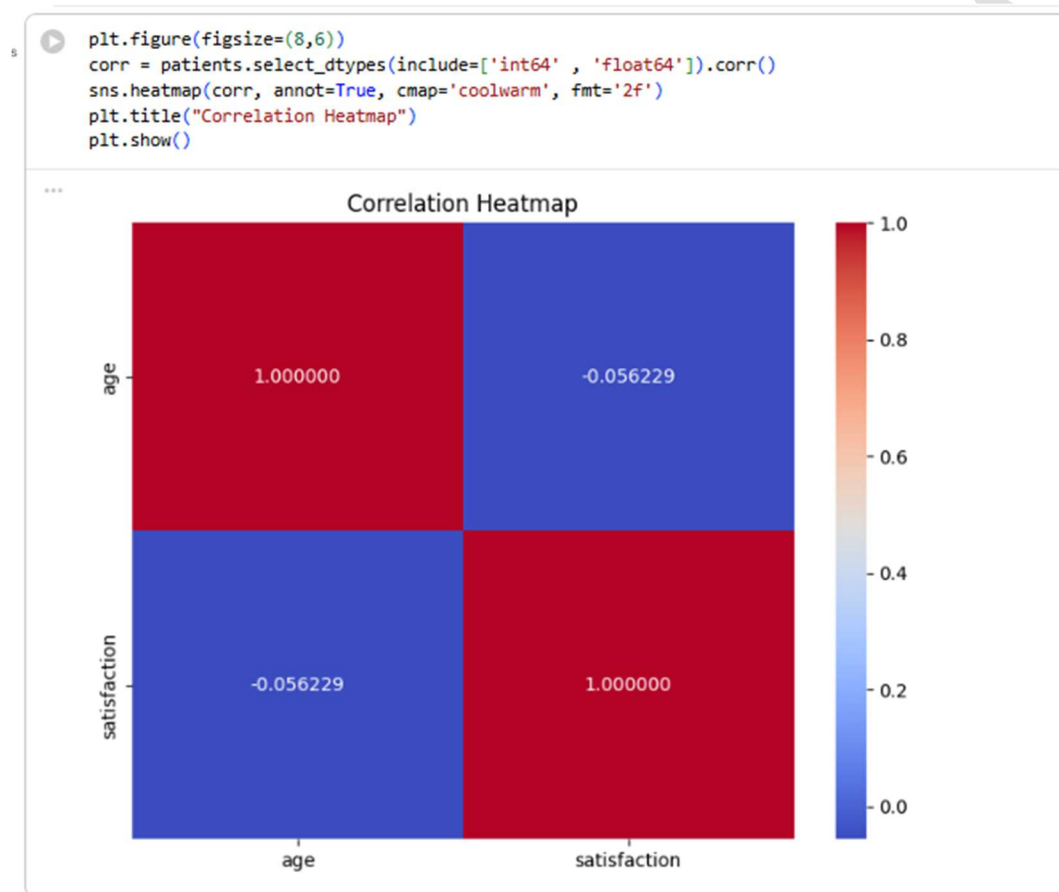
```
import matplotlib.pyplot as plt
import seaborn as sns

# List of numerical columns
num_cols = ['age', 'satisfaction']
for col in num_cols:
    plt.figure(figsize=(6,4))
    sns.histplot(patients[col], kde=True)
    plt.title(f"Distribution of {col}")
    plt.xlabel(col)
    plt.ylabel("Frequency")
    plt.show()
```



Q7. Create a heatmap to show the correlation between numerical variables.

Program screenshot:



Q8. Perform the following analysis on the satisfaction column:

- Find the average satisfaction score.
- Find the maximum and minimum satisfaction scores.
- Group by service and compare the average satisfaction scores.

Program screenshot:

```
15 1 # Find the target column
    2 target = None
    3
    4 # Based on info() and describe(), 'satisfaction' is the most likely target for analysis in this dataset.
    5 target = 'satisfaction'
    6
    7 print("\nTarget Column:", target)
    8
    9 print("\nAverage:")
   10 print(patients[target].mean())
   11
   12 print("\nMaximum:")
   13 print(patients[target].max())
   14
   15 print("\nMinimum:")
   16 print(patients[target].min())
   17
   18 ...
   19 Target Column: satisfaction
   20
   21 Average:
   22 79.597
   23
   24 Maximum:
   25 99
   26
   27 Minimum:
   28 60
```

Q9. Create a boxplot to show the distribution of patient satisfaction with respect to hospital service.

- Use `sns.boxplot()`.
- Plot service on the x-axis and satisfaction on the y-axis.
- Add a meaningful title and axis labels.
- Rotate the x-axis labels for better readability.

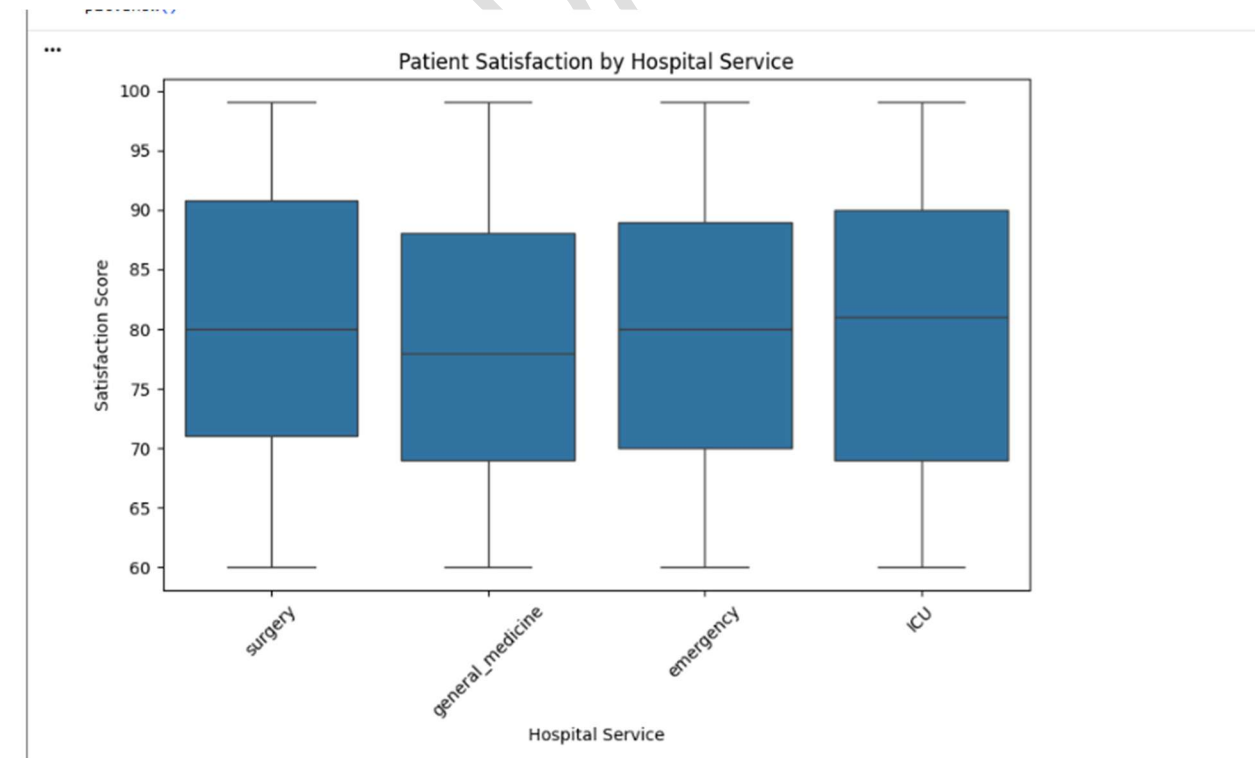
Program screenshot:

```
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(8,6))
sns.boxplot(x='service', y='satisfaction', data=patients)

plt.title("Patient Satisfaction by Hospital Service")
plt.xlabel("Hospital Service")
plt.ylabel("Satisfaction Score")

plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



Q10. Mini Project / Analysis Summary)

```
"""### Analysis Summary

- **Dataset:**
  The Patients dataset contains information about
  patient ID, name, age, arrival date,
  departure date, hospital service,
  and satisfaction scores.

- **Average Satisfaction Score:**
  The average patient satisfaction score is
  approximately **79.6%**, indicating that
  most patients are satisfied with the
  hospital services.

- **Patient Demographics:**
  Patients belong to different age groups,
  showing that the hospital serves a
  diverse population.

- **Most Active Department:**
  The department with the highest number
  of patient records is **(replace with your
  actual result)**.

- **Distribution Analysis:**
  Most patient satisfaction scores lie
  between **70% and 90%**, with only
  a few outliers.

- **Correlation Analysis:**
  The heatmap shows a very weak
  correlation between age and
  satisfaction.

- **Overall Observation:**
  The dataset is clean, well-structured,
  and suitable for Exploratory Data
  Analysis (EDA). It provides useful
  insights into patient demographics,
  hospital services, and patient
  satisfaction."""
```